

Tone Cooling Technology Co., Ltd

SPECIFICATION

nameofcommodity	Energy Storage 1P104S Cold Plate
Description	9. 03. 079. 045218079
roduct model	CP—00079
Product specification	2180*790*42
Rev.	A/0

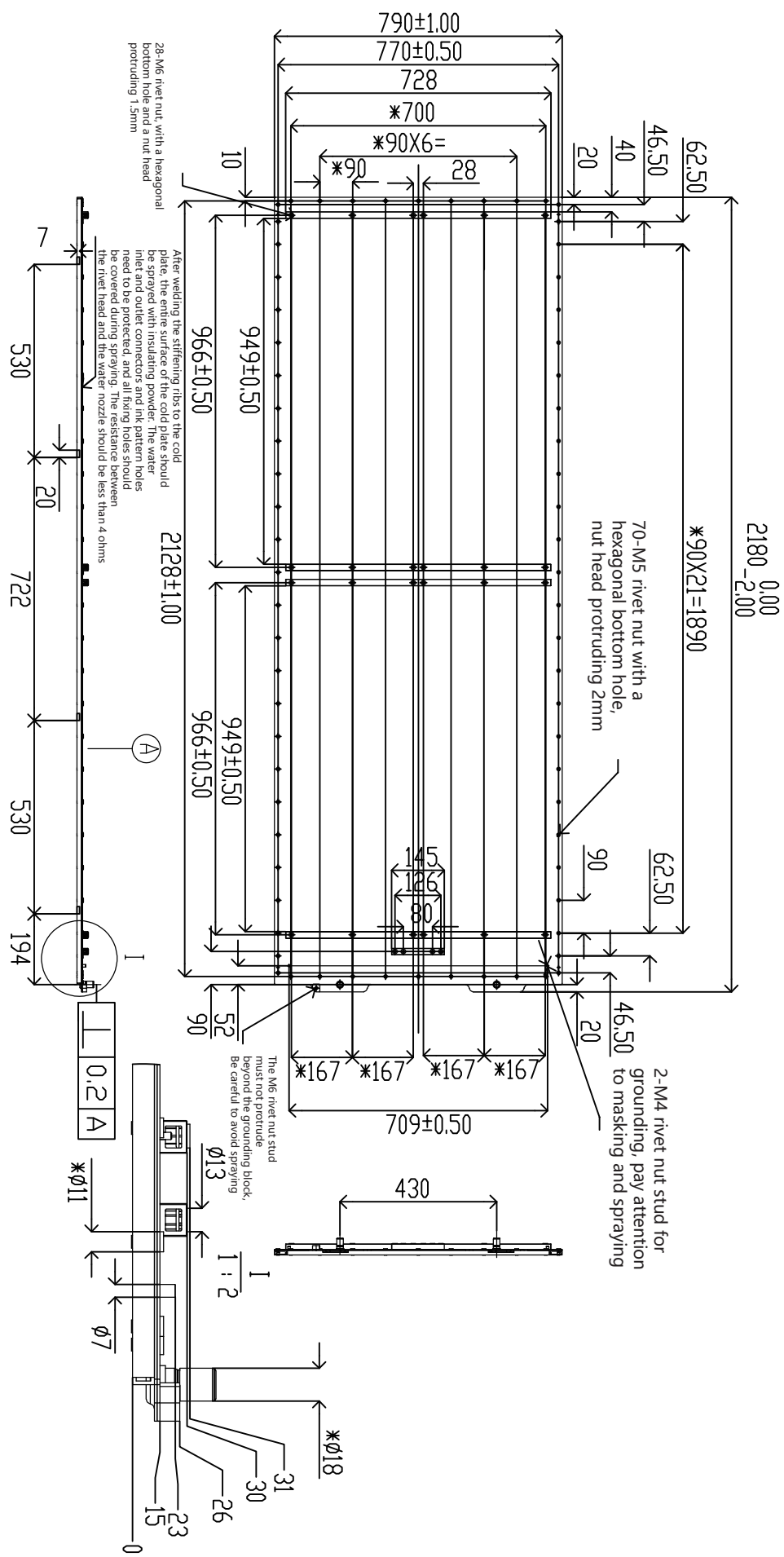
Factory Internal Approval Section

Sale Department	R&D Department	QA Department	Approval

Customer Confirmation Section

CM Department	QA Department	R&D Department	Approval

REV.	LTR.	DATE	UPDATE DESCRIPTION	REMARK	SIGN	APPROVER
A		2024/07/08	First Edition			
A		2024/08/20	Add a foaming process to cover the spraying position			



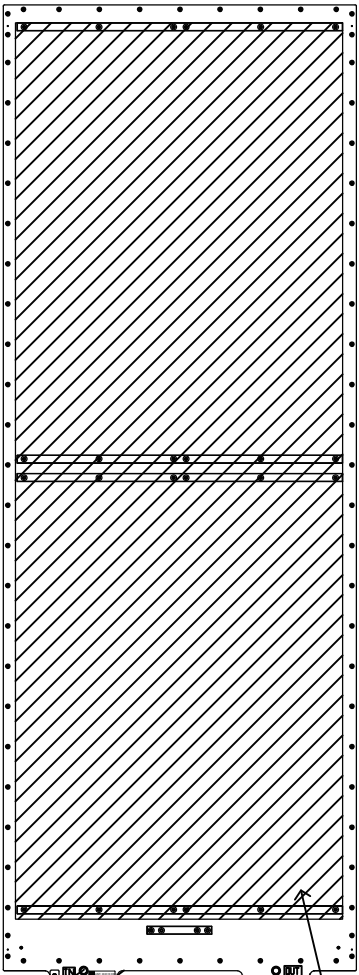
Technical Requirements:

1. Label: peel strength: 180°C, 12N.cm; the label and adhesive must remain free from defects such as lifting or adhesive overflow for 4 hours under conditions of minus 20°C and high temperature of 100°C;
 2. Process holes such as those for cold plate surface positioning, the pits after spraying must meet the requirement of being visually inconspicuous.
 3. Dimensions marked with an asterisk (*) are key control dimensions.
- For dimensions not specified, refer to 3D. For unstated tolerances, see GB/T1804-M. For unstated geometric tolerances, follow GB/T1184-K.

MATERIAL(SPEC)				QUANTITY/SEI		DESCRIPTION	
FINISH				SCALE		DRAWING NO.	
SELECT V				1:1		CP-00079	
DIM	TOL	A	B	C	D	UNIT	
-8	±0.03	±0.05	±0.10	±0.20		mm	
8-25	±0.05	±0.08	±0.20	±0.30			
25-80	±0.08	±0.12	±0.30	±0.40			
80-250	±0.12	±0.20	±0.40	±0.50			
250-	±0.20	±0.30	±0.50	±0.60			
				DATE		PART NO.	
				2024/08/20		9.03079045218079	
				DRAWING NO.		PAGE NO.	
				DRAWING NO.		1/1	
				CHECKED		APPROVED	
						REV	
						P1	

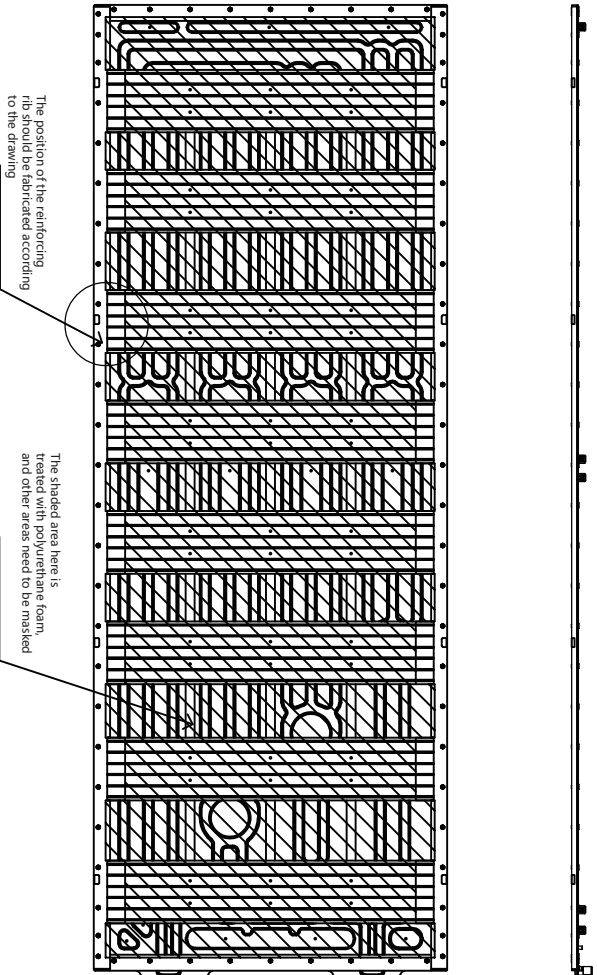
REV.	LTR	DATE	UPDATE DESCRIPTION	REMARK	SIGN	APPROVER
A		2024/07/08	First Edition			
A		2024/08/20	Add a foaming process to cover the spraying position			

The shaded area indicates the module installation zone. During painting, care should be taken to ensure there is no damage, roughness, scratching, protrusion, or foreign objects

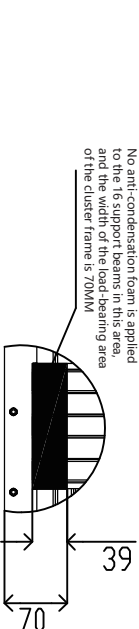


Factory information QR code area

Water inlet and outlet signs



Technical requirements: 1. Main material: The flat panel material is AL3003 MD composite layer 4343; the runner plate material is AL3003 MD; the bottom cross-support plate material is cold-rolled steel; the frame material is Q235; the 1P104S liquid-cooled panel flat panel is 1.5mm thick, the runner plate is 1.3-1.5mm thick, and the thinnest area of the runner plate after stamping should be 2.1mm. 2. Forming method: The main body of the cold plate is formed by brazing, and the main body of the product is formed by gluing and spot welding the cold plate and the reinforcing plate. There should be no visible defects such as welding slag, sharp corners or edges, no obvious scratches, rust, etc.; the appearance of the joint welding points requires a smooth surface, without welding slag, cold solder joint, or other defects; no weld beading, > flatness requirements, the flatness of the entire cold plate in its natural state should not exceed 3mm, and the flatness of the module area should not exceed 1.8mm; the flatness of the module installation area under a load of 0kg should not exceed 1mm. For dimensions not specified, refer to the 3D digital model. For dimensions not specified, the tolerance shall be implemented according to GB/T1804-2000M level. Aluminum connectors shall be protected with dust covers to prevent foreign objects from entering and damaging the connectors. Airtightness requirements: 100% airtightness testing is required for finished products. Xenon testing: test pressure is 250KPa±5KPa, test duration is 15S, leakage rate is paxM/52*10⁻³5MPa±5KPa. Inflation (inflation time depends on the equipment), pressure maintenance for 30S, test time for 60S. Water channel airtightness testing standard: the water-cooled board shall not deform, pressure drop shall be ≤3000, ambient temperature shall be 25°C, and under no environmental interference (such as wind or dust), the liquid-cooled board shall be placed in water, and 2.5bar compressed air shall be introduced. The testing equipment shall be used to pressurize the tested piece at a rate of 50KPa/s until the system breaks and stops. The failure pressure shall be recorded. F0.8Mpa: the force shall not be less than 3kN. The heat dissipation components of the ring part No. 2 in Wan 8*888 test number 17960512-2014 < * are required to meet standards such as "Prohibited Substances in Vehicles". Insulating powder with a thickness of 200+50um shall be applied. During spraying, the aluminum connectors shall be masked. The requirements are that the surface of the cold board shall have a DC voltage of 2700V, with a leakage current of <1mA for 60S; the resistance shall meet 21GQ (DC500V); the film shall not fall off, have no cracks, and the surface of the edge shall have no scratches; the insulating powder model is insulating epoxy resin powder material, color: black - touch area. The repair area of the module area shall use black insulating stickers with a diameter of 10mm. Only one repair point is allowed within the surrounding area, and the total number of repair points for the fog piece shall be 52. The distance between two adjacent repair points shall not exceed 10mm of the insulation diameter. Repair points shall not be sanded with sandpaper to avoid affecting the surrounding coating. The coating surface near the puncture point shall be cleaned to ensure no oil or impurities. Natural drying for 30S before use, and then paste a 10mm diameter repair patch to finally meet the insulation requirements; The 13h layer has a flame retardant rating of V0. It undergoes foaming treatment and has a thickness of T2mm. It is used for insulation and must not exhibit any signs of improper riveting or loose fastening. The failure torque is as follows: M6 hexagon nut 15Nm, M5 nut 7Nm, M4 nut 6Nm, DNM. Note: The hexagon pattern must meet the requirements of the go/no-go gauge. The liquid-cooled board must be printed with the factory information QR code. The coding principle is: company code + material code + production date + serial number. The inlet and outlet must be marked. The liquid-cooled board must not have any issues such as liquid leakage or seepage. The flow channels must not have cross-flow, and the liquid-cooled board must not show signs of rust (including nuts). The flow channels must be free of waste aluminum scraps, iron scraps, and other waste residues



No anti-condensation foam is applied to the top of the board heating area and the shape of the board heating area of the cluster frame is 70MM

The position of the reinforcing material to be fabricated according to the drawing

The shaded area here is treated with polyurethane foam, and other areas need to be masked

MATERIAL(SPEC)		QUANTITY/SE		DESCRIPTION		PAGE NO.	
FINISH		SCALE		DRAWING NO.		1/1	
1:1		CP-00079		9.03079.045218079		P1	
SELECT V		UNIT		CHECKED		APPROVED	
DIM TOL		DATE		DRAWING NO.		9.03079.045218079	
-8		+0.03 +0.05 +0.10 +0.20		2024/08/20		9.03079.045218079	
8-25		+0.05 +0.08 +0.20 +0.30		DRAWING NO.		9.03079.045218079	
25-80		+0.08 +0.12 +0.30 +0.40		CHECKED		APPROVED	
80-250		+0.12 +0.20 +0.40 +0.50		DRAWING NO.		9.03079.045218079	
250-		+0.20 +0.30 +0.50 +0.60		DRAWING NO.		9.03079.045218079	

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1P104S Parameter Description

Project	Parameter	Remarks
liquid cooling plate material	AL3003	
Bracket material	Q235	
Finished Size	2180*790*42	
Traffic	10L/Min	
Flow resistance	<25KPa	
Volume	1. 95L	
load-bearing	< 750KG	
Compatible battery cell	0. 5C 280AH/314AH	Customer selection
Quick-release adapter head	CQC14/ CQC16	Customer selection